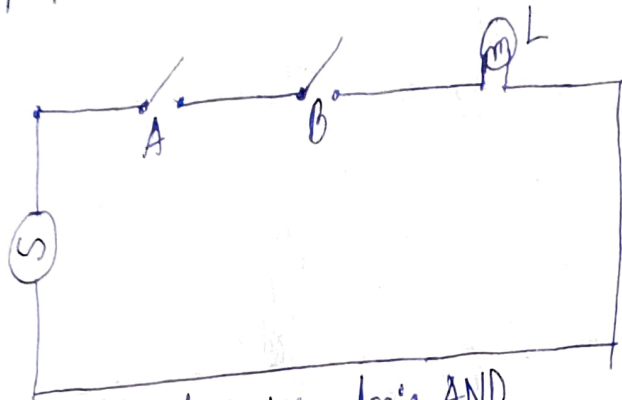


Table 1-6 Truth tables of logical operations

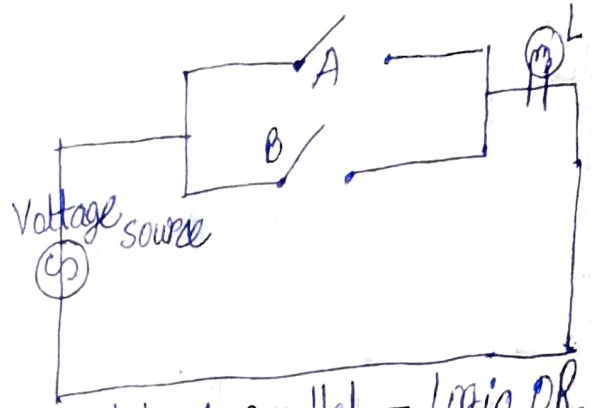
AND		
X	Y	$X \cdot Y$
0	0	0
0	1	0
1	0	0
1	1	1

OR		
X	Y	$X + Y$
0	0	0
0	1	1
1	0	1
1	1	1

NOT	
X	X'
0	1
1	0

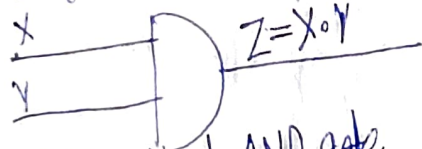


(a) Switches in series — logic AND

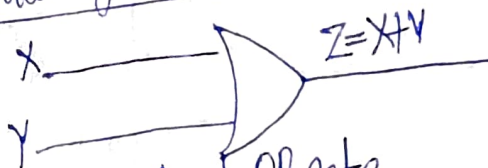


(b) Switches in parallel — Logic OR.

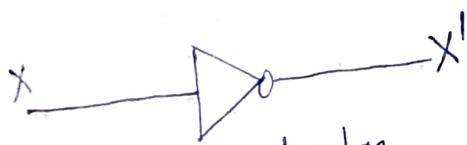
Figure 1.6: Symbols for digital logic circuits



(a) Two-input AND gate



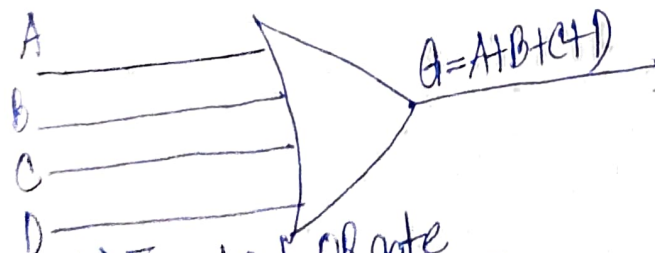
(b) Two-input OR gate



(c) NOT gate or inverter



(d) Three-input AND gate



(e) Four-input OR gate